



Scaleform CLIK

Scaleform CLIK

Prasad Silva, Matthew Doyle

2.0

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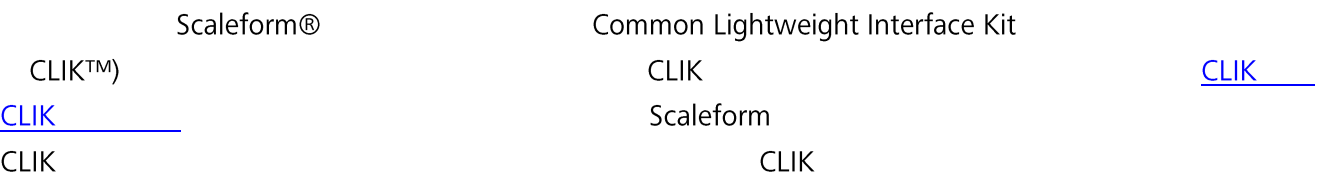
Document	Scaleform 4.0 CLIK AS3 User Guide
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

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1



1.1



1.1.1. Scaleform CLIK

UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropDownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

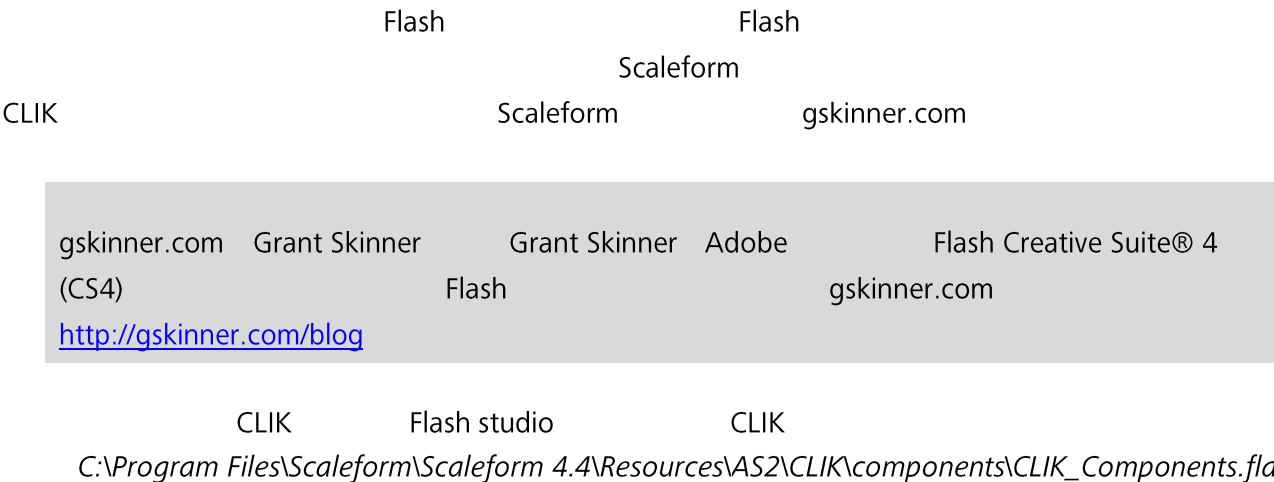
API

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
EventDispatcher	IList
Delegate	IListItemRenderer

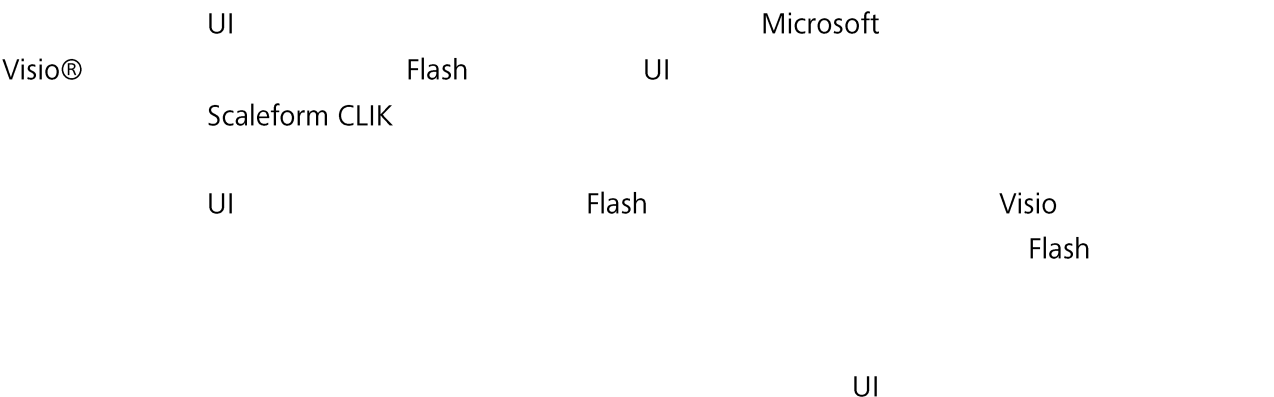
Constraints	GameDataProvider
-------------	------------------

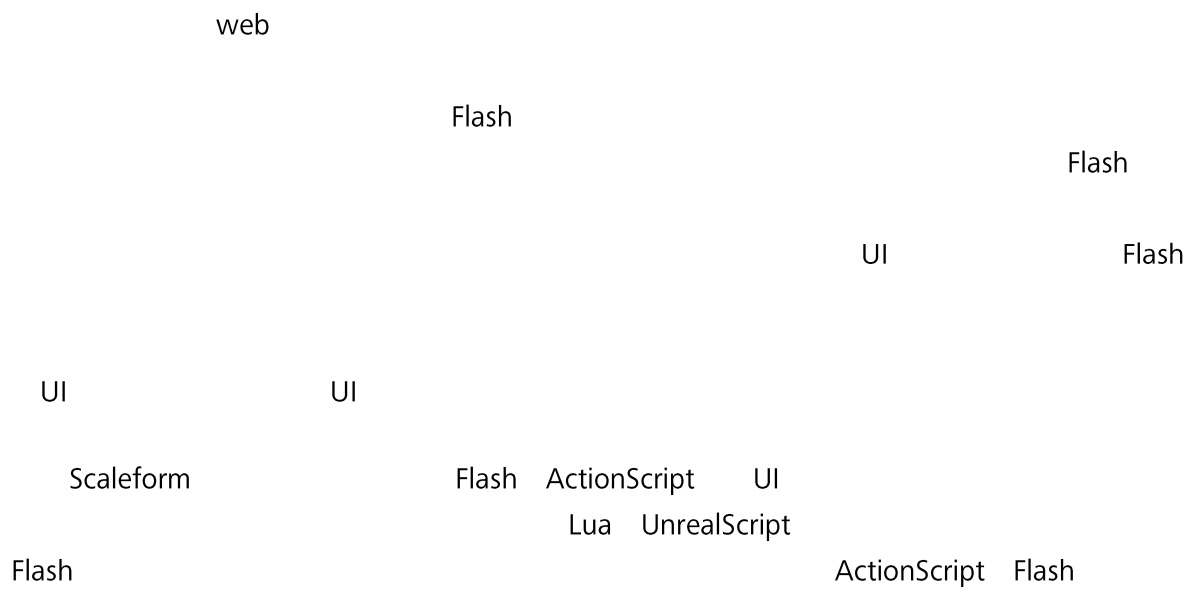
CLIK
CLIK
CLIK API
CLIK
CLIK Flash
CLIK

1.2



1.3 UI





2



- **User interaction**
- **Component setup** Flash
- **States**
- **Inspectable properties** Flash
- **Events** UI

- *Tips and tricks*

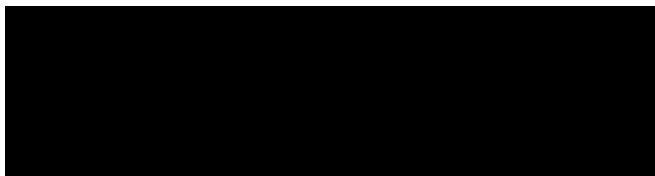
2.1

	Button	CheckBox	Label	TextInput			
TextArea		Button			CheckBox	Button	Label
	TextInput	TextArea					

1 *Free Realms*

2 *Free Realms* Check box

3 *Free Realms* Label



4: *Crysis Warhead* Text input

5	<i>Mercenaries 2</i>	Text area
---	-----------------------------	-----------

2.1.1 Button

6

Button CLIK

on gfx.controls.Button

textField

ScrollBar

Slider

Butt

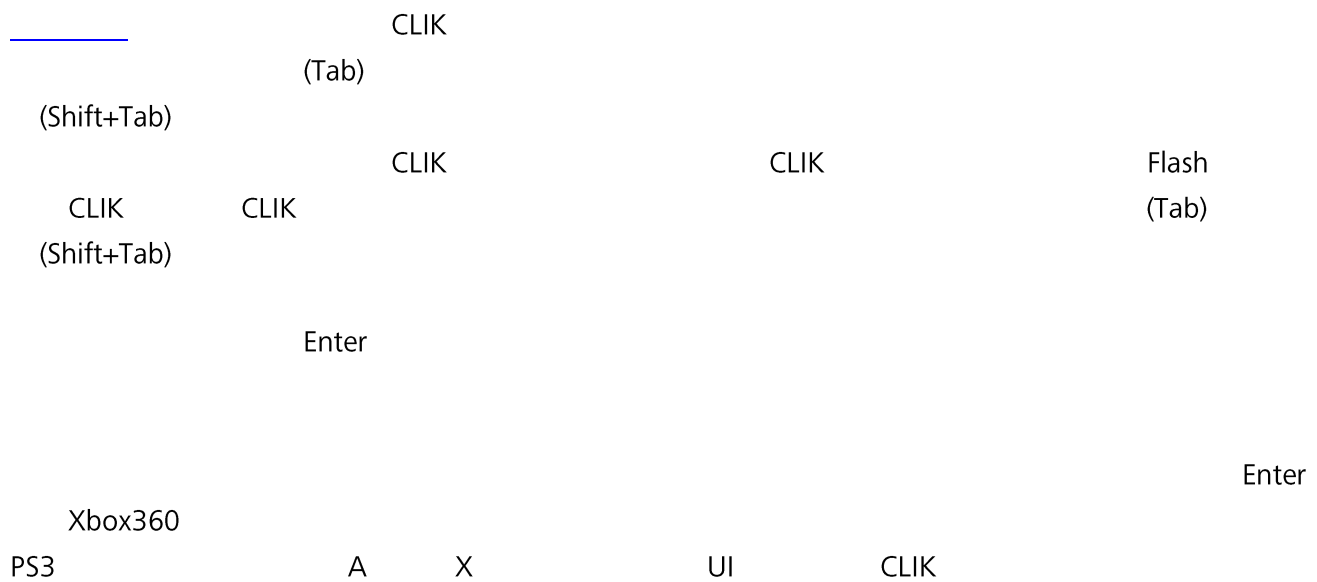
CLIK
Button

ToggleButton AnimatedButton

AnimatedToggleButton Button.fla

2.1.1.1

Programming



2.1.1.2

CLIK Button MovieClip

- **textField:** () TextField
- **focusIndicator:** () MovieClip

MovieClip
show hide over

Button

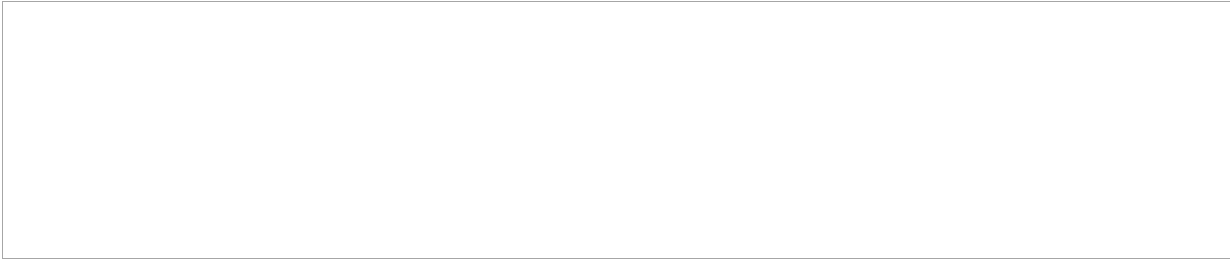
over

2.1.1.3

CLIK

- **up**

- *over*
- *down*
- *disabled*



8 Button

Flash

CLIK

2.1.1.4

Flash IDE Component

Inspector

Parameters

Component Inspector

CS4

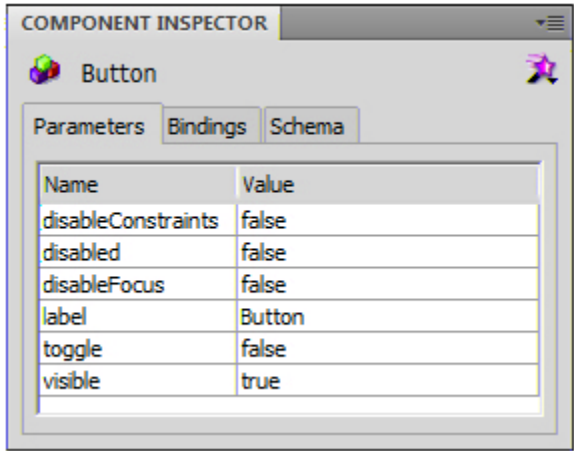
(Shift+F7)

AS

Window

Component

Inspector



9 CS4

:

disableConstraints	Button	textField
	true	

rollout	<i>mouseIndex</i>	0-3	
Press	<i>mouseIndex</i>	0-3	-
doubleClick	<i>mouseIndex</i>	<i>doubleClickEnabled</i>	true
	<i>mouseIndex</i>	0-3	-
Click	<i>mouseIndex</i>	0-3	-
dragOver	<i>mouseIndex</i>	0-3	-
dragOut	<i>mouseIndex</i>	0-3	-
releaseOutside	<i>mouseIndex</i>	0-3	-

ActionScript

```

myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:MouseEvent) {
    //
}

```

click myButton onButtonPress 函數。相同的代碼類型也可以用來處理其他的事件。事件控制碼中返回的Object參數（例子中名為 'event'）

CLIK

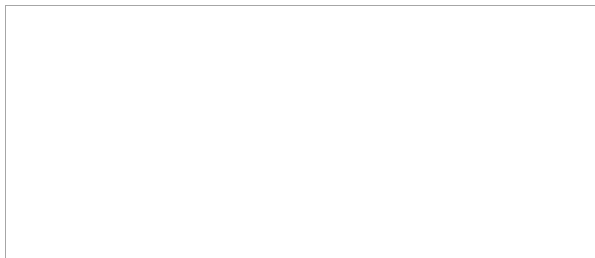
Toggle

```
btn.toggle = true;
```

```
btn.doubleClickEnabled = true;  
btn.addEventListener("doubleClick", this, "onDoubleClick");  
function onDoubleClick(e:Object):Void {  
    //  
    !  
}
```

```
btn.autoRepeat = true;
```

2.1.2 CheckBox



10

CheckBox

CheckBox

(gfx.controls.CheckBox)

true/false

ToggleButton

Toggle

2.1.2.1

CheckBox

2.1.2.2

CLIK CheckBox

MovieClip

- hide

CheckBox

- selected_down***

CLIK

ggle

textField /

Disabled	true	
Label		
Selected	true	CheckBox
Visible	false	

2.1.2.5

Object

- **type:**
- **target:**

CheckBox

Show		true			
Hide		false			
focusIn					
focusOut					
Select					
	<i>Selected</i>	true			
stateChange	<i>State</i>	String	up	over	down
	<u>CLIK</u>				
Rollover	<i>mouseIndex</i>				–
		0-3			
Rollout	<i>mouseIndex</i>				
		0-3			
Press	<i>mouseIndex</i>				–
		0-3			
doubleClick		<i>doubleClickEnabled</i>	true		
	<i>mouseIndex</i>				–
		0-3			
Click	<i>mouseIndex</i>				–
		0-3			

dragOver	<i>mouseIndex</i>	0-3	–
dragOut	<i>mouseIndex</i>	0-3	–
releaseOutside	<i>mouseIndex</i>	0-3	–

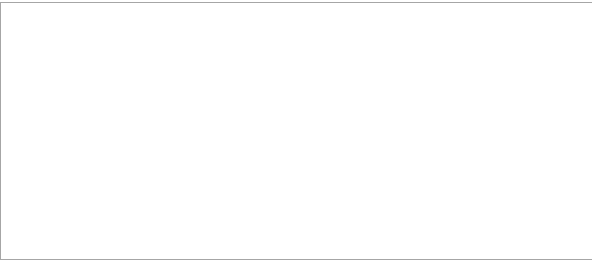
CheckBox Toggle

```

myCheckBox.addEventListener("select", this, "onCheckBoxToggle");
function onCheckBoxToggle(event:Object) {
    //
}

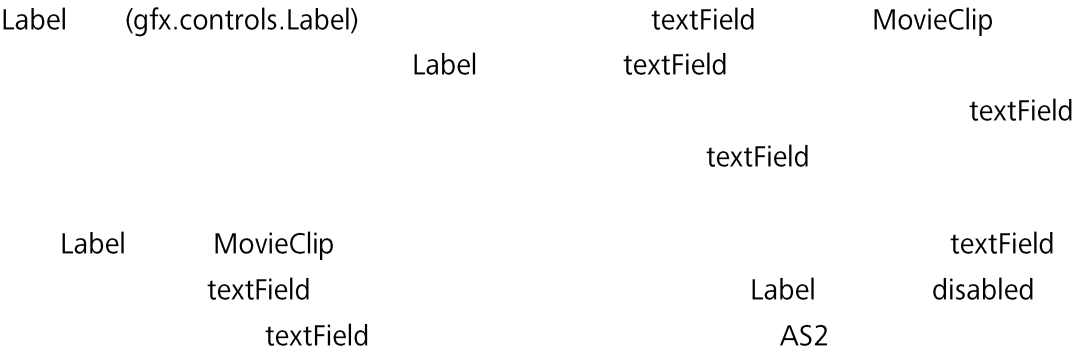
```

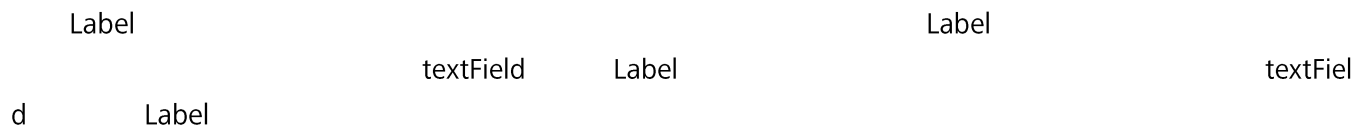
2.1.3 Label



12:

CLIK





2.1.3.1

Label

2.1.3.2

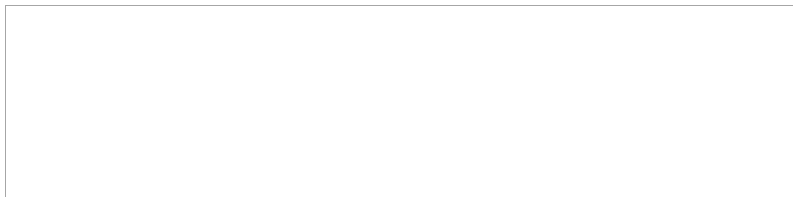
CLIK Label MovieClip

- ***textField:*** TextField Label

2.1.3.3

CLIK Label

- ***default*** enabled
- ***disabled***



13: Label

2.1.3.4

Label

text	
visible	false
disabled	true

2.1.3.5

Object

- ***type:***

- **target:**

Label

Show	true				
Hide	false				
stateChange	Label				
	State	Label	String	default	disabled

Label

```
myLabel.addEventListener("stateChange", this, "onStateChange");
function onStateChange(event:Object) {
    if (event.state == "default") {
        //
    }
}
```

2.1.3.6

Label

HTML

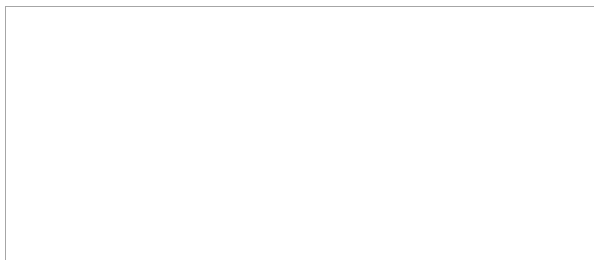
textField

HTML

Flash 8

```
lbl.htmlText = "<b>foo</b>bar";
```

2.1.4 TextInput



14: **TextInput.**

TextInput

(of control: TextInput)

textField

Label

textField

textField

HTML

TextInput textField

textField			Label	Label	
	textInput	textField	UI		PC
	textInput	disabled	textField		
TextInput	TextInput	TextInput	AS2	TextInput	TextInput
Studio	TextInput		TextInput		Flash
TextInput	textField,			HTML	
	(Shift+Arrows)	(Shift+Delete)			
(Ctrl+Insert)	(Shift+Insert)				
"	"		actAsButton		
			"over"	"out"	
actAsButton	"over"/"out"		"	"	"
	"	"			
	(0xAFFFFFFF)		"	"	"
"					

2.1.4.1

TextInput	textField	
		textField

2.1.4.2

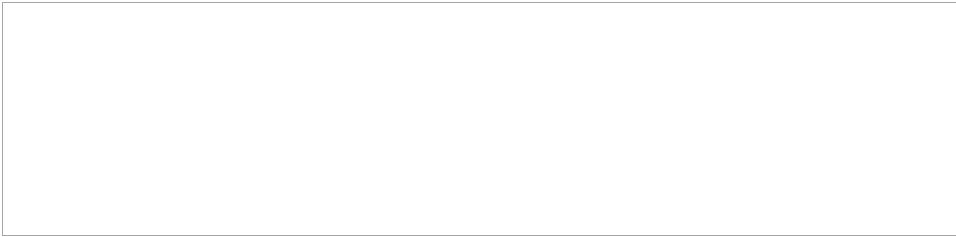
CLIK TextInput MovieClip

- **textField:** textField

2.1.4.3 States

CLIK TextInput disabled

- **default** enabled
- **focused** textField
- **disabled**



15: TextInput

2.1.4.4

TextInput :

text	textField		
visible	false		
disabled	true		
editable	false	TextInput	
maxChars	textField		
password	true	textField	'*'
	textField		
defaultText			
actAsButton	tab		

2.1.4.5

Object

- **type:**
- **target:**

TextInput

Show	true
Hide	false
focusIn	
focusOut	
textChange	textField
rollover	<i>mouseIndex</i> 0 3
rollout	<i>mouseIndex:</i> 0 3.

textField

```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
    //
}
```

2.1.4.6

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextInput

HTML

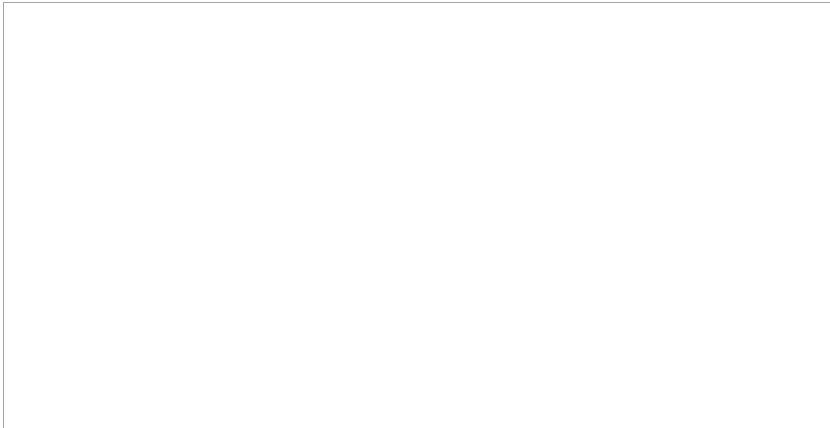
textField

HTML

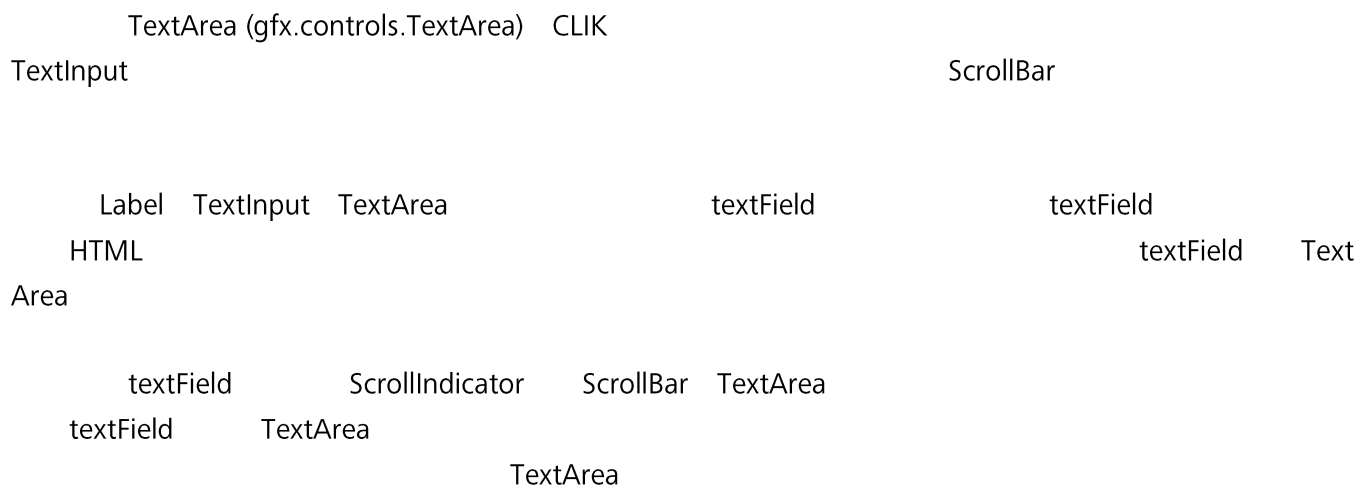
Flash 8

```
textInput.htmlText = "<b>foo</b>bar";
```

2.1.5 TextArea



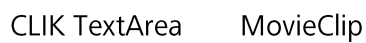
16: TextArea.



2.1.5.1



2.1.5.2

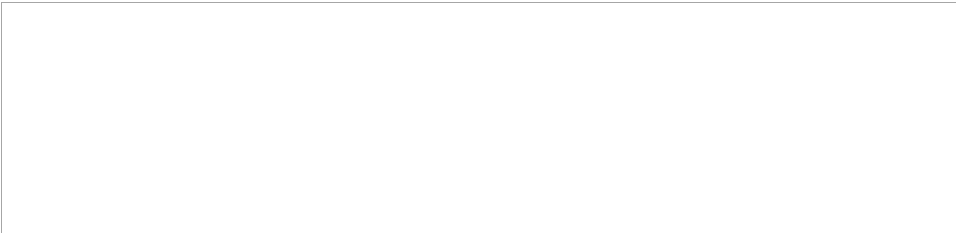


- **textField:** TextField

2.1.5.3

TextInput TextArea disabled

- **default** enabled
- **focused** textField
- **disabled**



17: TextArea

2.1.5.4

TextArea TextInput CLIK
ScrollBar 2.4

text	textField			
visible	false			
disabled	false			
editable	false	TextInput		
maxChars	.	textField		
scrollBar	CLIK ScrollBar TextArea	ScrollBar		ID
scrollPolicy	auto on ScrollBar ScrollBar	off ScrollBar		
defaultText				
actAsButton	tab			

2.1.5.5

Object

- **type:**
- **target:**

TextArea

Show	true
Hide	false
focusIn	
focusOut	
textChange	textField
Scroll	
rollover	mouseIndex 0 3
rollout	mouseIndex: 0 3.

TextArea

```
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    //
}
```

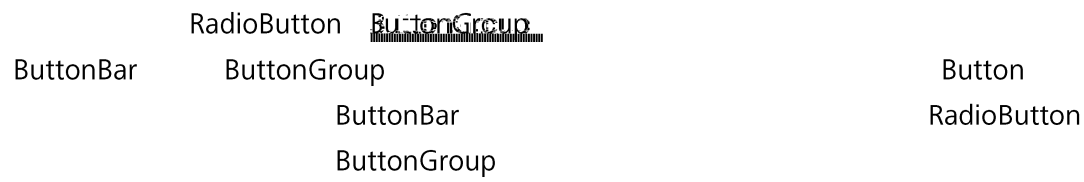
2.1.5.6

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextArea HTML textField HTML Flash 8

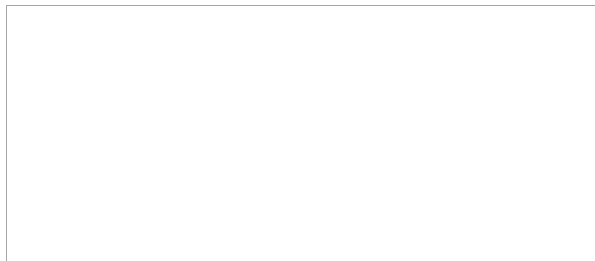
```
textInput.htmlText = "<b>foo</b>bar";
```

2.2

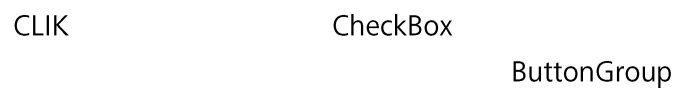
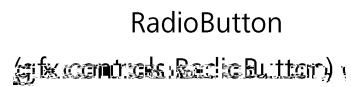


18: *Dawn of War II*

2.2.1 RadioButton



19: **RadioButton**



2.2.1.1

2.2.1.2

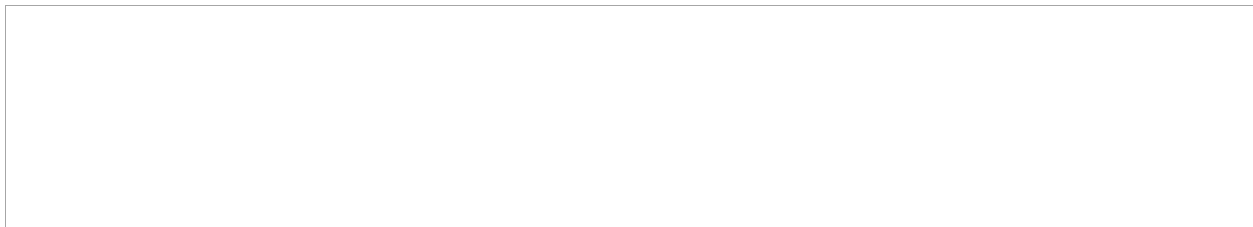
CLIK RadioButton MovieClip

- ***textField*** () TextField
- ***focusIndicator*** MovieClip MovieClip
MovieClip show hide

2.2.1.3

CheckBox

- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***
- ***selected_down***
- ***selected_disabled***



20: **RadioButton**

RadioButton Button
RadioButton CLIK

2.2.1.4

Button

RadioButton

disableF

ocus

Toggle

Label	Button				
Visible	false				
Disabled	true				
disableConstraints	Button	textField			
		true			t
	extField				disa
	bled textField		textField /		
Selected	true		RadioButton		
Data					RadioButt
	on				
Group	ButtonGroup			RadioButton	
		ButtonGroup			
		<i>_root</i>		<i>_root</i>	

2.2.1.5

Object

- *type:*
- *target:*

RadioButton

Show	true				
Hide	false				
focusIn					
focusOut					
Select					
	Selected	true			
stateChange					
	State	String	up	over	down
	<u>CLIK</u>				

rollover	<i>mouseIndex</i>	0-3	-
rollOut	<i>mouseIndex</i>	0-3	-
Press	<i>mouseIndex</i>	0-3	-
doubleClick	<i>mouseIndex</i>	<i>doubleClickEnabled</i>	true
	<i>mouseIndex</i>	0-3	-
Click	<i>mouseIndex</i>	0-3	-
dragOver	<i>mouseIndex</i>	0-3	-
dragOut	<i>mouseIndex</i>	0-3	-
releaseOutside	<i>mouseIndex</i>	0-3	-

RadioButton

```
myRadio.addEventListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
    //
}
```

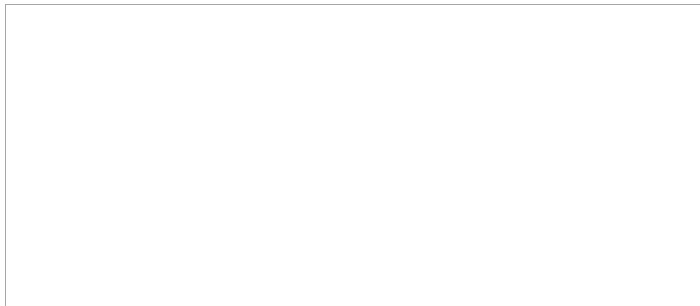
2.2.1.6

RadioButton

```
import gfx.controls.ButtonGroup;
var myGroup:ButtonGroup = new ButtonGroup("groupName");
radio1.group = myGroup;
radio2.group = myGroup;
```

```
someMovie.radio1.group = myGroup;
someMovie.radio2.group = myGroup;
```

2.2.2 ButtonGroup



21: ButtonGroup.

CLIK ButtonGroup
(gfx.controls.ButtonGroup)

CLIK
up

CheckBox _

RadioButton

ButtonGro

2.2.2.1

RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

Object

- **type:**
- **target:**

ButtonGroup

change			
• <i>item</i>	CLIK	Button	
• <i>data</i>		AS2	
itemClick			
• <i>item</i>	CLIK	Button	

ButtonGroup

```
myGroup.addEventListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
    if (event.item == myRadio) {
        //
    }
}
```

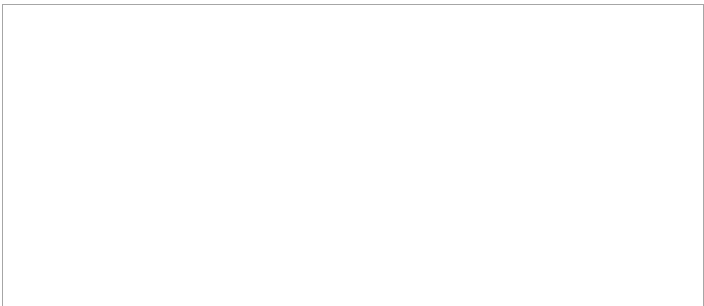
2.2.2.6

```
var selectedRadio:Button = group.selectedButton;
```

ButtonGroup

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```

2.2.3 ButtonBar



22: ButtonBar.

ButtonBar
(gfx.controls.ButtonBar) ButtonGroup dataProvider
dataProvider tab UI

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1

ButtonGroup



CLIK ButtonBar MovieClip

2.2.3.3

CLIK

2.2.3.4

Flash
IDE ButtonBar

Visible	false	ButtonBar
Disabled	true	buttonBar
itemRenderer	ID	

Direction	
Spacing	<i>direction</i>
autoSize	true
buttonWidth	autoSize true

2.2.3.5

Object

- **type:**
- **target:**

ButtonBar

Show	true
Hide	false
focusIn	
focusOut	
change	• <i>index</i> 0 1 • <i>renderer</i> CLIK Button • <i>item</i> dataProvider AS2 Object • <i>data</i> dataProvider AS2 Object
itemClick	• <i>index</i> 0 1 • <i>item</i> dataProvider AS2 Object • <i>data</i> dataProvider AS2 Object • <i>mouseIndex</i> 0-3

```

myBar.addEventListener("itemClick", this, "onItemClick");
function onItemClick(event:Object) {
    processData(event.data);
    //
}

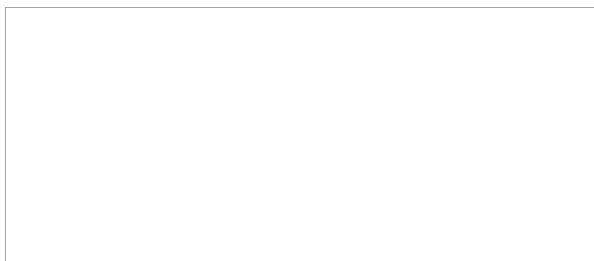
```

2.3

Slider ScrollIndicator ScrollBar ScrollBar Slider
 ScrollIndicator

23: *Mercenaries 2*

2.3.1 ScrollIndicator



24: **ScrollIndicator.**

CLIK ScrollIndicator
(gfx.controls.ScrollIndicator) textField
 ID TextArea

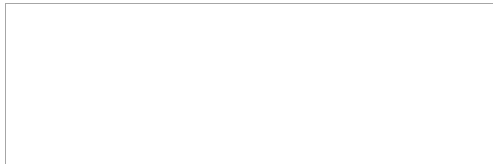
2.3.1.1

2.3.1.2

CLIK ScrollIndicator MovieClip

- ***thumb***: CLIK Button
- ***track*** MovieClip

2.3.1.3



25: **ScrollIndicator**

2.3.1.4

scrollTarget	TextArea	textField	
	textField		ScrollIndicator
Visible	false		
Disabled	false		
offsetTop			
offsetBottom			

2.3.1.5

Object

- ***type***:
- ***target***:

ScrollIndicator

Show	true
Hide	false
Scroll	• <i>position</i>

```
mySI.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

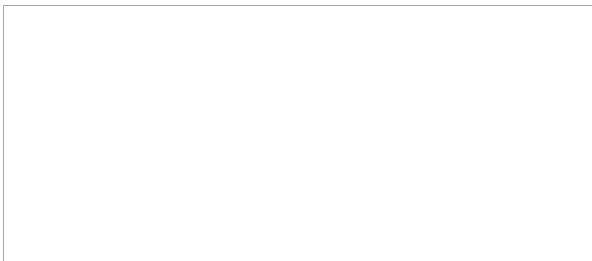
2.3.1.6

```
scrollInd.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollInd.positon = currPos;
```

_rotation

```
scrollInd.direction = "horizontal";
```

2.3.2 ScrollBar



26: **ScrollBar.**

CLIK ScrollBar(gfx.controls.ScrollBar)

2.3.2.1

ScrollBar

(Shift)

trackMode

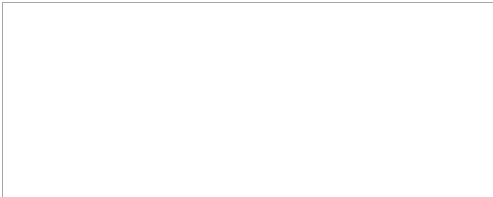
trackMode

2.3.2.2

CLIK ScrollBar MovieClip

- *upArrow* CLIK
- *downArrow*: CLIK
- *thumb*: CLIK
- *track*: CLIK

2.3.2.3



27: ScrollBar

ScrollBar ScrollIndicator

2.3.2.4

scrollTarget	TextArea	textField
trackMode	scrollPage	
	scrollToCursor	
Visible	false	
Disabled	false	
offsetTop		
offsetBottom		

2.3.2.5

Object

- **type:**
- **target:**

ScrollBar

show	true
hide	false
scroll	• <i>position</i>

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

2.3.2.6

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollBar.positon = currPos;
```

_rotation

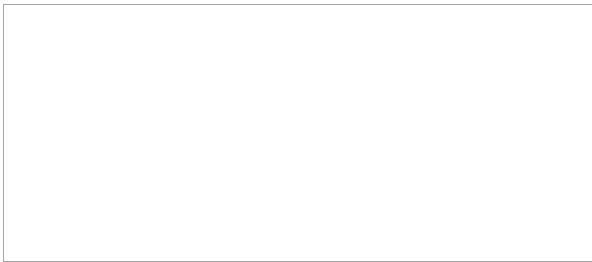
```
scrollBar.direction = "horizontal";
```

scrollPage

1

```
scrollBar.trackScrollPageSize = pageSize;
```

2.3.3 Slider



28: **Slider.**

Slider(gfx.controls.Slider)

2.3.3.1

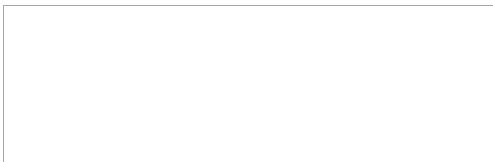
home end

2.3.3.2

CLIK Slider MovieClip

- ***thumb*** CLIK
- ***Track*** CLIK

2.3.3.3



2.3.3.4

Slider

Visible	false
----------------	-------

Disabled	false	
Value	Slider	
Minimum		
Maximum		
Snapping	true	
snapInterval		false
liveDragging	true	Slider
offsetLeft		
offsetRight		

2.3.3.5

Object

- **type:**
- **target:**

Slider

show	true
hide	false
focusIn	
focusOut	
change	Slider

Slider

```
mySlider.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("slider value: " + event.target.value);
    //
}
```

2.4

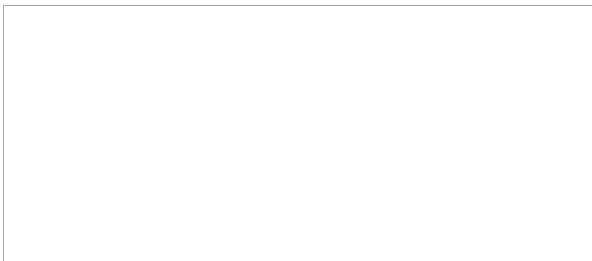
CLIK	NumericStepper	OptionStepper	ScrollingList	TileList	
DropdownMenu		NumericStepper		DataProvider	ListItemR
enderer		ScrollingList	TileList		
NumericStepper	OptionStepper			ScrollingList	
TileList			ScrollIndicator		
ScrollBar	DropdownMenu				ScrollingList
TileList					

30: ***Mercenaries 2***

31: ***Halo Wars*** 'Difficulty Settings'

32: *Mass Effect* 'Display Setting'

2.4.1 NumericStepper



33: **Numeric Stepper.**

NumericStepper
(gfx.controls.NumericStepper)

2.4.1.1

NumericStepper

(Home)

(End)

2.4.1.2

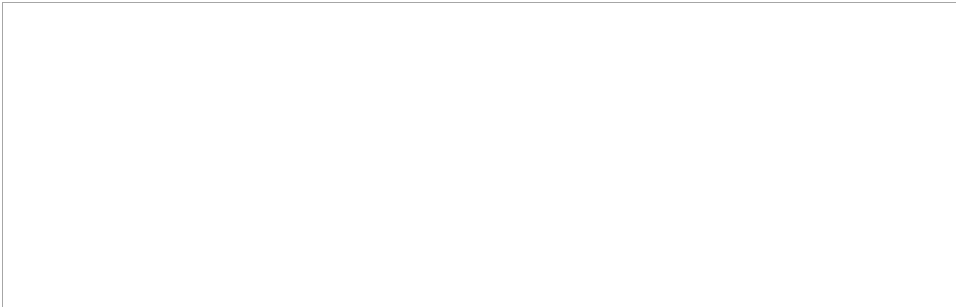
CLIK NumericStepper MovieClip

- ***textField***: TextField
- ***nextBtn***: CLIK Button
- ***prevBtn***: CLIK Button

2.4.1.3

NumericStepper

- **default** enabled
- **focused** textField
- **disabled**



34: NumericStepper

2.4.1.4

NumericStepper MovieClip

visible	false
disabled	false
value	NumericStepper.
minimum	NumericStepper
maximum	NumericStepper

2.4.1.5

Object

- **type:**
- **target:**

NumericStepper

Show	true
Hide	false
Change	NumericStepper
stateChange	NumericStepper
• <i>state</i>	String "default" "focused" "disabled"

NumericStepper

```
myNS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("ns value: " + event.target.value);
    //
}
```

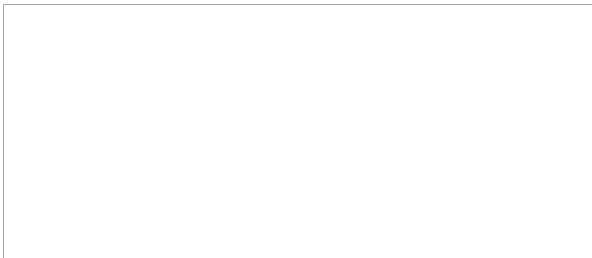
2.4.1.6

/

```
ns.stepSize = 0.5;
```

```
ns.labelFunction = function(value:Number) {
    switch(value) {
        case 1:
            return value + "st";
        default:
            return value + "th";
    }
}
```

2.4.2 OptionStepper



35: OptionStepper.

OptionStepper
(gfx.controls.OptionStepper), NumericStepper
dataProvider

OptionStepper

dataProvider

```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1

NumericStepper

OptionStepper

2.4.2.5

Object

- *type:*
- *target:*

OptionStepper

Show	true			
Hide	false			
Change	OptionStepper			
stateChange	OptionStepper			
	• state	String	"default"	"focused"
	"disabled"			

OptionStepper

```
myOS.addListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("os value: " + event.target.selectedItem);
    //
}
```

2.4.3 ListItemRenderer

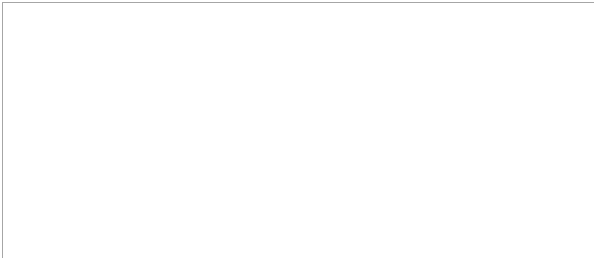
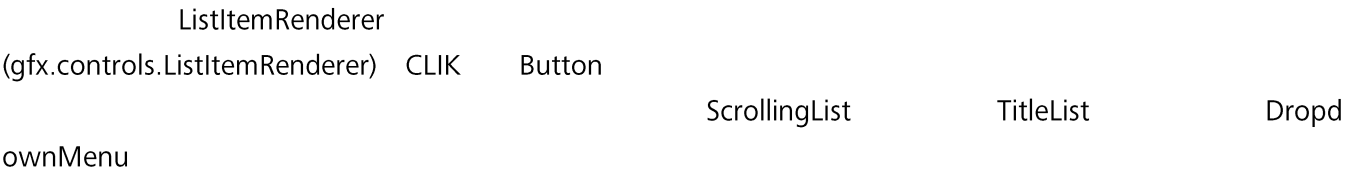


图 1: ListItemRenderer.



2.4.3.1

2.4.3.4

ListItemRenderer

Label	ListItemRenderer
Visible	false
Disabled	true

2.4.3.5

Object

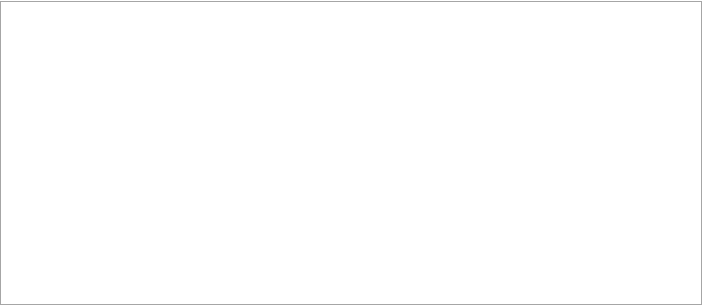
- *type:*
- *target:*

ListItemRenderer

Show	true				
Hide	false				
focusIn					
focusOut					
Select	• <i>Selected</i> true				
stateChange	State	String	up	over	down
	CLIK				
Rollover	• <i>mouseIndex</i> - 0-3				
Rollout	• <i>mouseIndex</i> 0-3				
Press	• <i>mouseIndex</i> - 0-3				
doubleClick	<i>doubleClickEnabled</i> true				
	• <i>mouseIndex</i> -				

		0-3	
Click	• <i>mouseIndex</i>	-	
		0-3	
dragOver	• <i>mouseIndex</i>	-	
		0-3	
dragOut	• <i>mouseIndex</i>	-	
		0-3	
releaseOutside	• <i>mouseIndex</i>	-	
		0-3	

2.4.4 ScrollingList



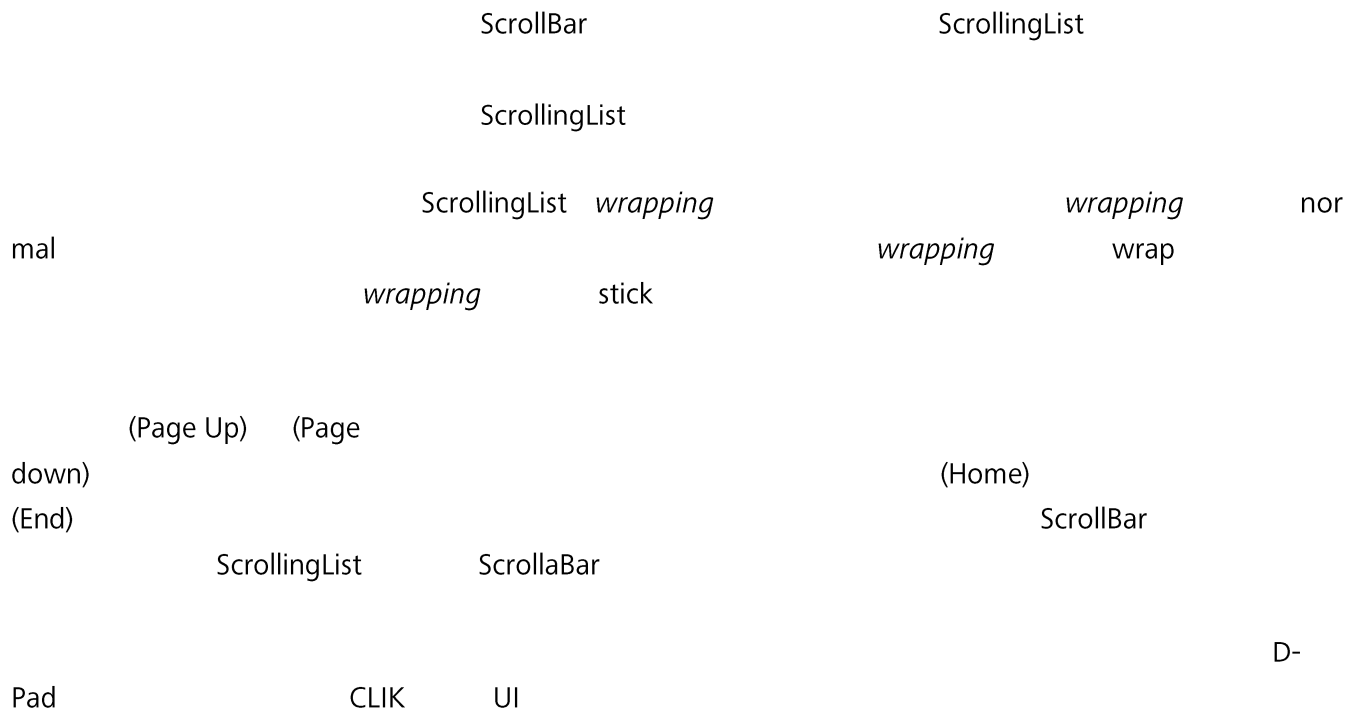
39: **ScrollingList.**

ScrollingList
 (gfx.controls.ScrollingList)
 ScrollIndicator
 ScrollBar
 dataProvider

```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4"];
```

crollingList
 FLA
 ListItemRenderer

2.4.4.1



2.4.4.2

`ScrollingList` `ScrollingList`

2.4.4.3

`ScrollingList`

- **default** enable
- **focused**
- **disabled**

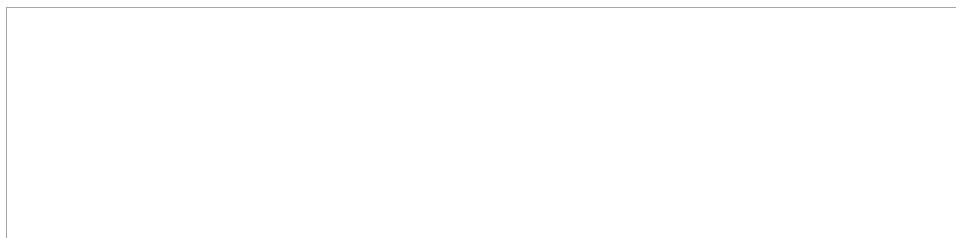


图40: `ScrollingList`

2.4.4.4

ScrollingList	MovieClip
Visible	false
Disabled	true
itemRenderer	ListItemRenderer <i>rendererInstanceName</i>
rendererInstanceName	ScrollingList r r1 r2 r3 1
Scrollbar	ScrollBar ScrollingList ScrollingList
Margin	<i>rendererIns tanceName</i>
rowHeight	<i>rendererInstanceName</i>
paddingTop	<i>rendererInstanceName</i>
paddingBottom	<i>rendererInstanceName</i>
paddingLeft	<i>rendererInstanceName</i>
paddingRight	<i>rendererInstanceName</i>
thumbOffsetTop	
thumbOffsetBottom	
thumbSizeFactor	1.0 1.0

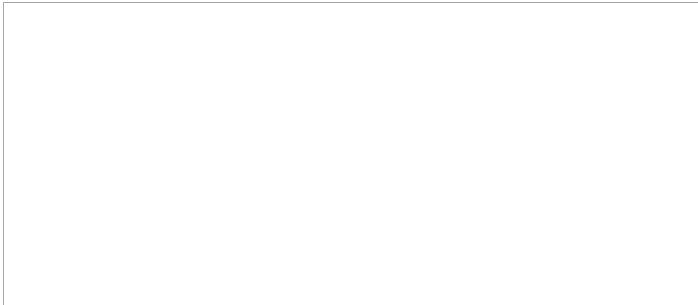
2.4.4.5

Object

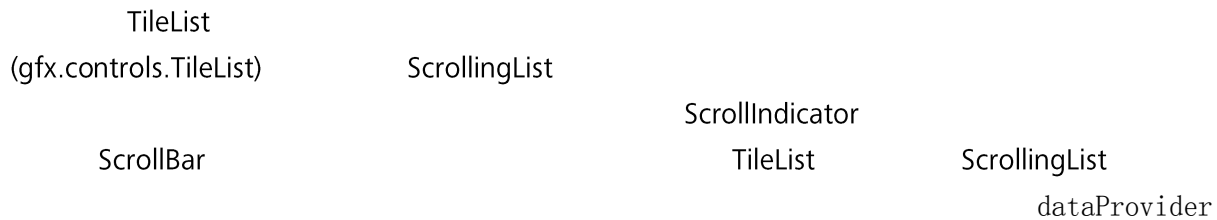
- *type:*


```
//
//      labelFunction
list.labelFunction = function(itemObj:Object):String {
    //
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

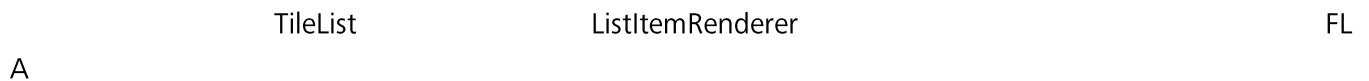
2.4.5 TileList



41: TileList.



```
tileList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```



2.4.5.1.



(Page Up)

(Page down)

(End)

TileList

ScrollBar

(Home)

ScrollBar

2.4.5.2

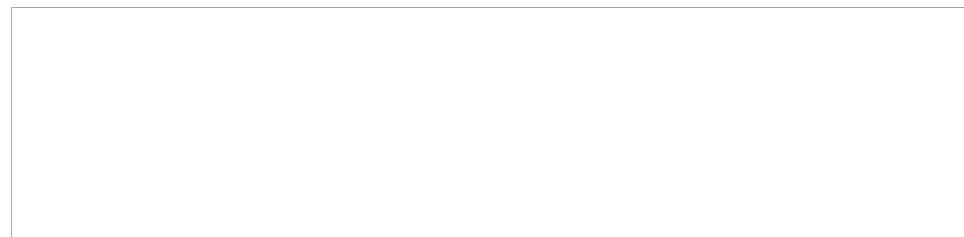
TileList

TileList

2.4.5.3

TileList

- **default** enable
- **focused**
- **disabled**



42:

TileList

2.4.5.4

TileList

MovieClip

Visible	false
Disabled	true
itemRenderer	ListItemRenderer rendererInstanceName
rendererInstanceName	TileList r

		1	r1	r2	r3
scrollbar		ScrollBar			
		TileList			
		TileList			
Margin					<i>rendererIn</i>
		<i>stanceName</i>			
rowHeight					<i>rendererInstanceName</i>
columnWidth					<i>rendererInstanceName</i>
externalColumnCount					<i>rendererInstanceName</i>
		TileList			
Direction					

2.4.5.5

Object

- *type:*
 - *target:*
- TileList

Show		true		
Hide		false		
focusIn				
focusOut				
Change				
	• <i>index</i>		0	1.
itemPress				
	• <i>renderer</i>	CLIK		
	• <i>item</i>			dataProvider
	AS2			
	• <i>index</i>	dataProvider		
			0	1
	• <i>mouseIndex</i>			

0 3				
itemClick				
• <i>renderer</i>		CLIK		
• <i>item</i>				dataProvider
AS2				
• <i>index</i>	dataProvider			
		0		1
• <i>mouseIndex</i>				
	0 3			
itemDoubleClick				
• <i>renderer</i>		CLIK		
• <i>item</i>				dataProvider
AS2				
• <i>index</i>	dataProvider			
		0		1
• <i>mouseIndex</i>				
	0 3			
itemRollOver				
• <i>renderer</i>		CLIK		
• <i>item</i>				dataProvider
AS2				
• <i>index</i>	dataProvider			
		0		1
• <i>mouseIndex</i>				
	0 3			
itemRollOut				
• <i>renderer</i>		CLIK		
• <i>item</i>				dataProvider
AS2				
• <i>index</i>	dataProvider			
		0		1
• <i>mouseIndex</i>				
	0 3			

TileList

```

myList.addEventListener("focusIn", this, "onListFocused");
function onListFocused(event:Object) {
    trace("tile list was focused!");
    //
}

```

2.4.5.6

```

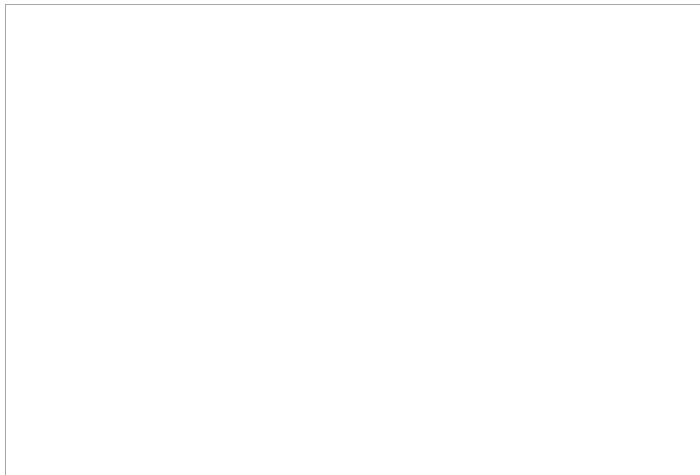
// ScrollingList          'label'
//
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

//
//
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

//
//
// labelFunction
list.labelFunction = function(itemObj:Object):String {
    //      Logic to construct a label
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

2.4.6 DropdownMenu



43: DropdownMenu.

DropdownMenu
(gfx.controls.DropdownMenu)
DropdownMenu
rollBar ScrollIndicator dataProvider ScrollingList TileList Sc
dataProvider dataProvider dataProvider DropdownMenu

```
dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];
```

DropdownMenu DropdownMenu ListItemRenderer
DropdownMenu ListItemRenderer FLA
ScrollIndicator DropdownMenu ScrollBar
DropdownMenu

2.4.6.1

DropdownMenu (Enter)
ScrollBar ScrollingList TileList
DropdownMenu

2.4.6.2

DropDownMenu

MovieClip

- **textField** () TextField
- **focusIndicator**: ()
MovieClip
show hide

2.4.6.3

DropDownMenu

ToggleButton CheckBox

- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***
- ***selected_down***
- ***selected_disabled***

44: DropdownMenu

DropdownMenu
DropdownMenu

Button

CLIK

2.4.6.4

DropDownMenu

Visible	false
---------	-------

Disabled	false			
Dropdown	ScrollingList			
	TileList	DropDownMenu		
dropdownWidth	-			
	1			
itemRenderer				
scrollbar				
margin				
paddingTop				
paddingBottom				
paddingLeft				
paddingRight				
thumbOffsetTop				
thumbOffsetBottom				
thumbSizeFactor	thumb	1.0	thumb	1.0
		1.0	thumb	
offsetX				
offset				
Extent	offsetX	dropdownWidth	-	
	1			
Direction				

2.4.6.5

Object

- *type:*
- *target:*

DropDownMenu

change

Button

Show	true								
Hide	false								
focusIn									
focusOut									
Change	<i>index</i>01. <i>data</i>AS2								
Select	<i>Selected</i>true								
stateChange	State	String	up	over	down				
	CLICK								
rollover	<i>mouseIndex</i>0-3								
rollout	<i>mouseIndex</i>0-3								
Press	<i>mouseIndex</i>0-3								
doubleClick	<i>doubleClickEnabled</i>true <i>mouseIndex</i>0-3								
Click	<i>mouseIndex</i>0-3								
dragOver	<i>mouseIndex</i>0-3								
dragOut	<i>mouseIndex</i>0-3								
releaseOutside	<i>mouseIndex</i>-								
	0-3								

2.4.6.6

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
    if (e.target.isOpen) {
        // open
    }
}
```

DropDownMenu

```
//          ;
//          /          FLA
//          DropdownMenu ScrollingList
//  ScrollBar .
attachMovie("DropDownMenu", "dd", this.getNextHighestDepth(),
            {dropdown: "ScrollingList"});
dd.dataProvider = ["one", "two", "three", "four", "five", "six"];

//
//
//
onEnterFrame = function() {
    dd.dropdown.scrollBar = "ScrollBar";
    onEnterFrame = null;
}
```

```
// ScrollingList          'label'
//
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

//
//

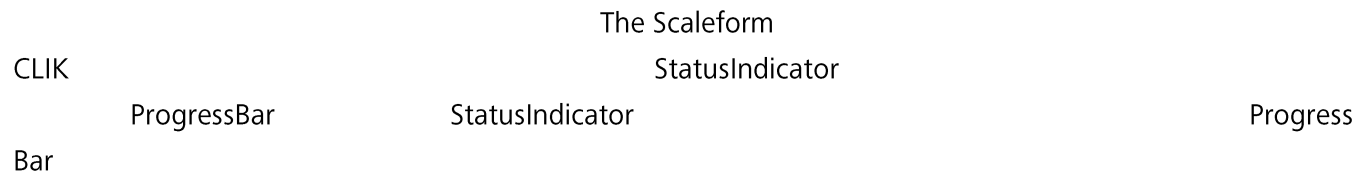
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];
//
//      labelFunction
list.labelFunction = function(itemObj:Object):String {
//
```

```

}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

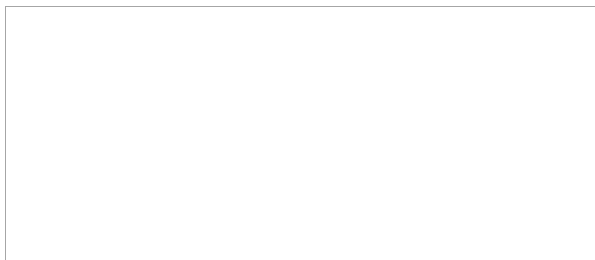
```

2.5



45: *Mercenaries 2* Faction

2.5.1 StatusIndicator



46: **StatusIndicator.**

StatusIndicator component
(gfx.controls.StatusIndicator)

2.5.1.1

StatusIndicator

2.5.1.2

CLIK

StatusIndicator

MovieClip

stop()

gotoAn

dStop()

2.5.1.3

StatusIndicator

2.5.1.4

StatusIndicator

MovieClip

visible	false
disabled	false
Value	
minimum	
maximum	

2.5.1.5

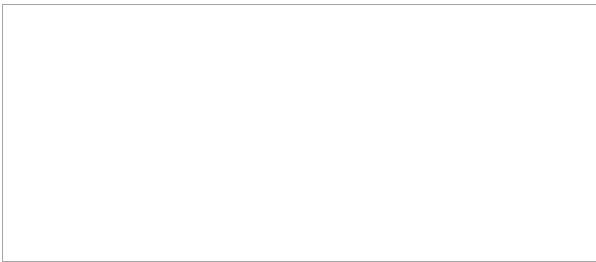
Object

- **type:**
- **target:**

StatusIndicator

Show	true
Hide	false

2.5.2 ProgressBar



47: ProgressBar.

ProgressBar
(gfx.controls.ProgressBar) StatusIndicator

b

bytesLoaded bytesTotal

2.5.2.1

ProgressBar

2.5.2.2

StatusIndicator CLIK StatusIndicator MovieClip

ProgressBar stop()

ProgressBar gotoAndStop()

2.5.2.3

ProgressBar

2.5.2.4

ProgressBar MovieClip

visible	false		
disabled	false		
target	ProgressBar	bytesLoaded	bytesTotal
mode	properties.	ProgressBar	manual
	polled	setProgress	bytesLoaded

bytesTotal	event	progress	byt
esLoaded	bytesTotal		

2.5.2.5

Object

- **type:**
- **target:**

ProgressBar

Show	true
Hide	false
progress	ProgressBar
complete	

```

myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
    if (event.type == "progress") {
        //
    } else
    {
        trace("Loading complete!");
    }
}

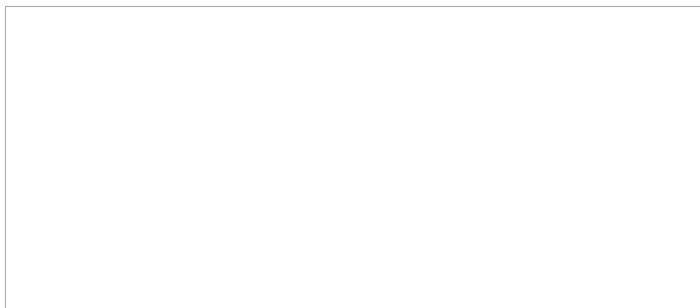
```

2.6

Scaleform
CLIK
UI
Dialog
UILoader
ViewStack
Dialog
ViewStack
UI
UILoader

48: *Halo Wars* Dialog

2.6.1 Dialog



49: Dialog

CLIK
Dialog
(gfx.controls.Dialog)

MovieClip

MovieClip

Dialog.show()

2.6.1.1

Dialog

2.6.1.2

CLIK Dialog MovieClip :

- **closeBtn** () CLIK Button
- **cancelBtn** () CLIK Button cancel
- **submitBtn** () CLIK Button OK
- **dragBar** () MovieClip

2.6.1.3

Dialog MovieClip

2.6.1.4

Dialog MovieClip

Visible	false
Disabled	false

2.6.1.5

Object

- **type:**
- **target:**

Dialog

Show	true
Hide	false
submit	Dialog
• data	Dialog
	getSubmitData

	etSubmitData	AS2	true	g
Close	Dialog			

```

myDialog.addEventListener("submit", this, "onLoginSubmit");
function onLoginSubmit(event:Object) {
    trace("Received data from dialog: " + event.data);
}

```

2.6.2 UILoader

CLIK UILoader
 (gfx.controls.UILoader)
 Scaleform SWF/GFX UILoaders

2.6.2.1

UILoader SWF/GFX UILoader

2.6.2.2

CLIK UILoader MovieClip :

- **bg** () MovieClip
 UILoader

2.6.2.3

UILoader SWF/GFX UILoader

2.6.2.4

UI UILoader MovieClip

Visible	false		
autoSize	true	UILoader	
maintainAspectRatio	true	UILoader	false
		UILoader	
Source	SWF/GFX		

Timeout			UIMLoad
	er	ioError	

2.6.2.5

Object

- *type:*
- *target:*

UIMloader

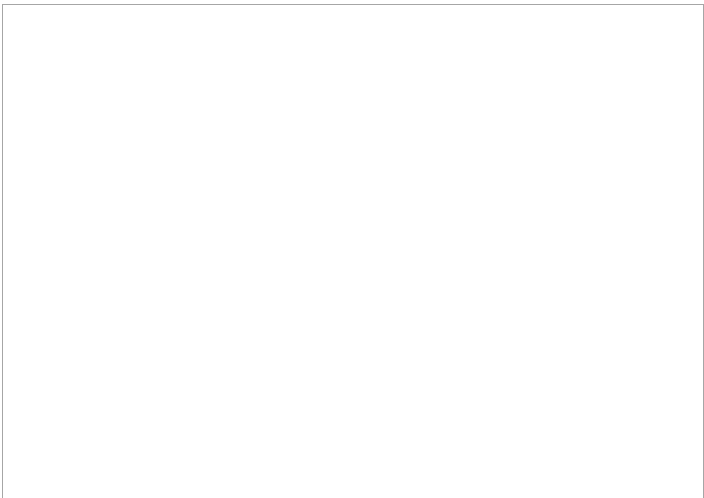
Show	true		
Hide	false		
progress		a	b
loaded	0	100	
complete			
ioError			

```

myUIMloader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
    displayErrorMessage("Could not load" + event.target.source);
}

```

2.6.3 ViewStack



50: ViewStack

CLIK

ViewStack

(gfx.controls.ViewStack)

x

Accordion

RadioButton

ViewStack

TabBo

2.6.3.1

ViewStack

ViewStack

2.6.3.2

MovieClip

CLIK

ViewStack

2.6.3.3

ViewStack

ViewStack

2.6.3.4

ViewStack

MovieClip

Visible	false
Cache	true
	ViewStack
	targetGroup

targetGroup	ButtonGroup	change	ID	Radio
Buttons		Flash IDE		ID

2.6.3.5

ViewStack

3

Scaleform

CLIK

3.1

Scaleform CLIK

3.1.1

CLIK

Flash

- 1. Flash
- 2.
- 3. Flash
- 4.
- 5.
- 6. OK

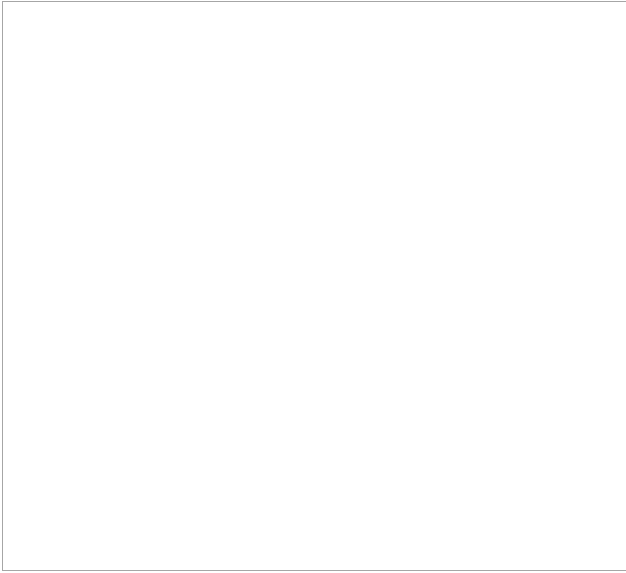
1px

51: Grid

SWF

2

2



52:

3.1.2

Flash

Scaleform

Flash

Photoshop®

PNG

3.1.3

—

3.1.4

Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- ~~64x64~~
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash

Scaleform CLIK

3.2.1

Flash

3.2.1.1 1

FLA

CLIK

FLA

1. *CLIK_Components.fla*

2. FLA

Copy F

LA *Paste*

3. FLA *Rename* Button

4. FLA *Properties*

5. *Identifier* 3

6. *Paste*

3.2.1.2 **2**

1. *library* *Properties*
2. *Properties* gfx.controlsButton *Class* (CTRL+C)
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export* *ActionScript*
6. *Class* (CTRL+V) textField
7. *OK*
8. *library* *Component Definition*
9. *Class* textField (CTRL+V) textField
10. *OK*

53:

54:

3.3

Scaleform
CLIK

CLIK

CLIK

- Flash
- Flash
- Photoshop

Flash

Illustrator®

FLA

Flash

[CLIK](#)

3.3.1 StatusIndicator



55: **StatusIndicator.**

 StatusIndicator

indicator

track *Track*

Indicator

 Photoshop

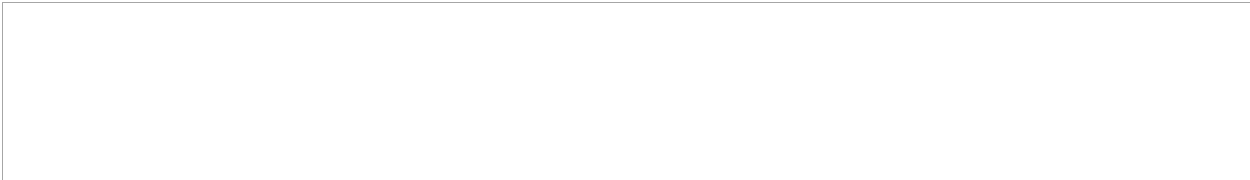


56: **StatusIndicator PSD**

57 Photoshop StatusIndicator PSD

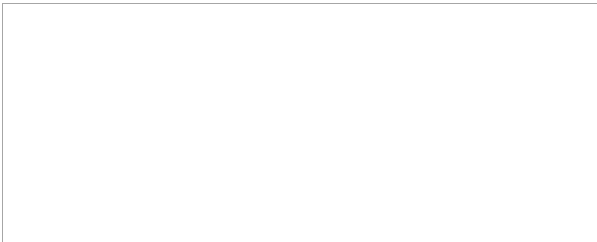
1. Photoshop StatusIndicator
2. PSD
3. Flash *File Import -> Import to Stage*
4. StatusIndicator PSD
5. *Import Convert layers to:* Layers
6. OK
7. Flash PSD 10 1 10 1 10 10
layer 1
- 8.
- 9.
10. *layer 1* 6
11. *Insert layer 1* 11
12. *layer 2 (Ctrl+X)*
13. *layer 1* 11 *(Ctrl+Shift+V) layer 1*
14. 10 *(layer*
1) 5 *layer*
2 11 *layer 3* 16 *layer*
4 21 10 PSD 51
15. 2-10

16. Remove Frames 55
17. *indicator layer*
18. *track* *track*
19. *layer 1* *indicator*
20. *indicator* *track* *actions layer* *track* *indicator*



58:

21. StatusIndicator
22. 1 10
- 23.
- 24.



59: StatusIndicator

3.4

Scaleform CLIK

3.4.1

Scaleform

SWF Scaleform

Flash UI

3.4.2

Flash Scaleform Scaleform Scaleform textFidld Slate
Mobile Slate CJK ASCII Sla
Mobile CJK
te Mobile
m Scaleform textFields Scaleform 3.4.4 Scalefor
UI

3.4.3 textField

textFields
textField
Embed (Window > Property F
LA OK FLA

3.4.4

Scaleform
SWF SWF Scaleform
Scaleform textField
gfxfontlib fla swf
Scaleform
fontconfig.txt Scaleform

fontconfig.txt

Scaleform

\$TITLE

\$TITLE=Scaleform

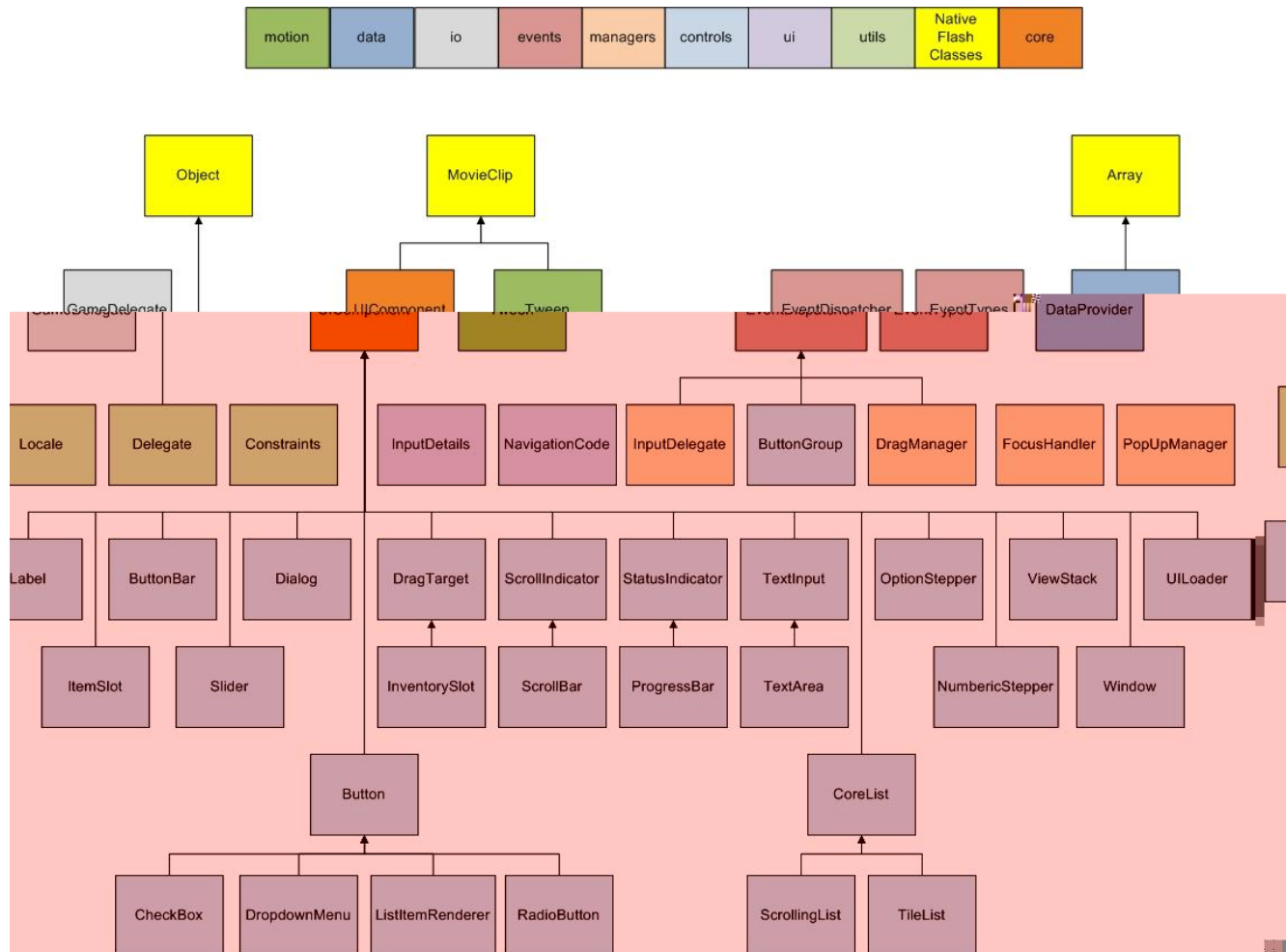
Scaleform

Scaleform

Scaleform

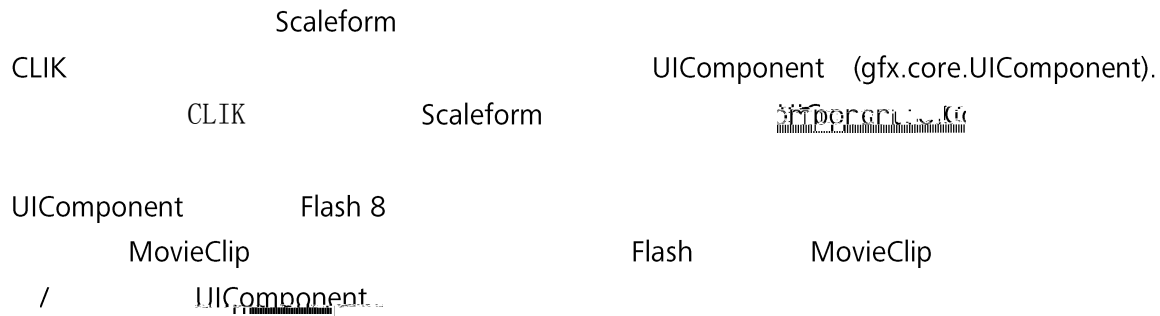
4

Scaleform CLIK



60: Scaleform CLIK

4.1 UI *UIComponent*



- ***disabled;***
- ***visible;***
- ***focused;***
- ***width;***
- ***height;***
- ***displayFocus:***

true

UIComponent

- ***configUI:***
- ***draw:***
- ***changeFocus:***
- ***scrollWheel:***

UIComponent EventDispatcher

4.1.1

- ***onLoad()***
- ***MovieClip***
- ***configUI()***
- ***validateNow()***
- ***draw()***

4.2

Scaleform
CLIK

4.2.1

ListItemRenderer RadioButton
CheckBox ScrollBar Button
Button (gfx.controls.Button) Button
CLIK Button
CLIK

- ***up***
 -
 - ***down***
 - ***disabled***
- over***

Button CLIK Button
selected_

- ***up***
-
- ***down***
- ***disabled***
- ***selected_up***
-
- ***selected_over***
- ***selected_down***
- ***selected_disabled***

CLIK

[CLIK](#)

CLIK getStatePrefixes()

```
private function getStatePrefixes():Array {  
    return (_selected) ? ["selected_", ""] : [""];  
}
```

CLIK

selected_

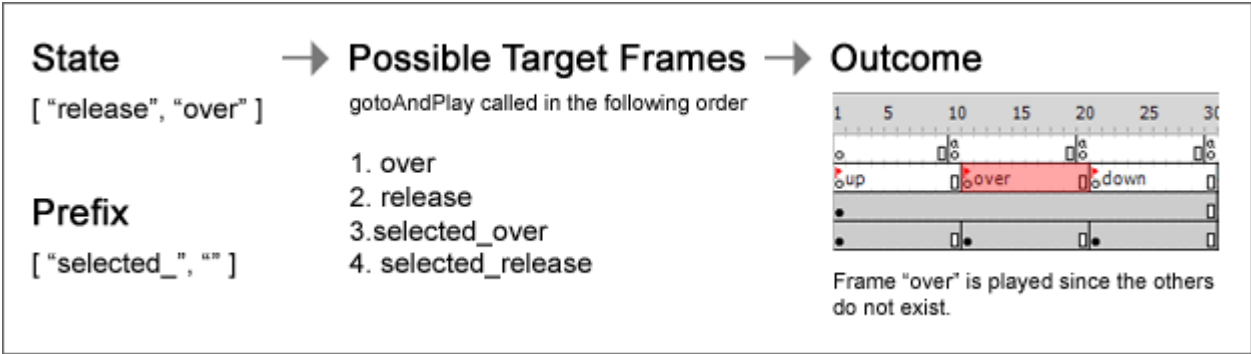
getStatePrefixes()

stateMap

CLIK

```
private var stateMap:Object = {
    up:["up"],
    over:["over"],
    down:["down"],
    release: ["release", "over"],
    out:["out","up"],
    disabled:["disabled"],
    selecting: ["selecting", "over"],
    kb_selecting: ["kb_selecting", "up"],
    kb_release: ["kb_release", "out", "up"],
    kb_down: ["kb_down", "down"]
}
```

getStatePrefixes()



61:

4224

CLIK Scrolli

ngList OptionStepper Slider
TextArea

- *default*
- *focused*
- *disabled*

4.2.3

- *default*
- *disabled*

4.2.4

				StatusIndicator			ProgressBar	
							50	
0	10	5	50%	gotoAndStop()	25	50	50%	
			updateValue()					

4.3

Scaleform CLIK
model

event

ActionScript 2

EventDispatcher (gfx.events.EventDispatcher)
EventDispatcher.initialize()
super()

API

UIComponent

CLIK
CLIK
UIComponent
EventDispatcher
Eve

ntDispatcher

4.3.1

4.3.1.1

addEventListener()
removeEventListener()

String

```
EventTypes (gfx.events.EventTypes)
(EventTypes.SHOW      show  )
```

```
buttonInstance.addListener("itemClick", this, "callBack");
function callBack(eventObj:Object):Void {
    buttonInstance.removeListener("itemClick", this, "callBack");
}
```

itemClick

callback

CLIK

4.3.1.2

CLIK dispatchEvent()

```
dispatchEvent({type:"itemClick", item:selectedItem});
```

4.4

Scaleform

CLIK

Scaleform

A

ctionScript

Selection.setFocus

FocusHandler (gfx.managers.FocusHandler)

FocusHandler

4.4.1

CLIK

Tab

true

```
Selection.setFocus(firstComponentOrMovieClip);
```


4.5

Scaleform CLIK Flash 8
 MovieClip CLIK MovieClip

4.5.1

InputDelegate class
 (gfx.managers.InputDelegate) InputDelegate InputDetails (gfx.u
 i.InputDetails) InputDelegate.readInput()
 InputDetails InputDelegate

InputDetails

- key
-
-

keyUp

keyDown

- navEquivalent(navigation equivalent)
 "right" NavigationCode

"up" "down" "left"

gfx.managers.InputDelegate

FocusHandler InputDelegate

the handleInput()

InputDetails pathToFocus



62: handleInput()

FocusHandler handleInput() false null navEquivalent Sca
leform

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {  
    if (details.navEquivalent == "left")  
    { //      NavigationCode.LEFT  
        doSomething();  
        return true;  
    }  
    return false;  
}
```

pathToFocus true fa
lse

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {  
    var nextItem:MovieClip = pathToFocus.shift();  
    var handled:Boolean = nextItem.handleInput(details, pathToFocus);  
    if (handled) { return true; }  
    //  
    return false; //      true  
}
```

DropDownMenu handleInput

pathToFocus

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var handled:Boolean = dropdown.handleInput(details);
    if (handled) { return true; }
    //
    return false; //          true
}
```

InputDelegate.instance
FocusHandler

```
InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
    var details:InputDetails = event.details;
    if (details.value = Key.TAB) { doSomething(); }
}
```

InputDelegate

W A S D

InputDelegate

Nintendo

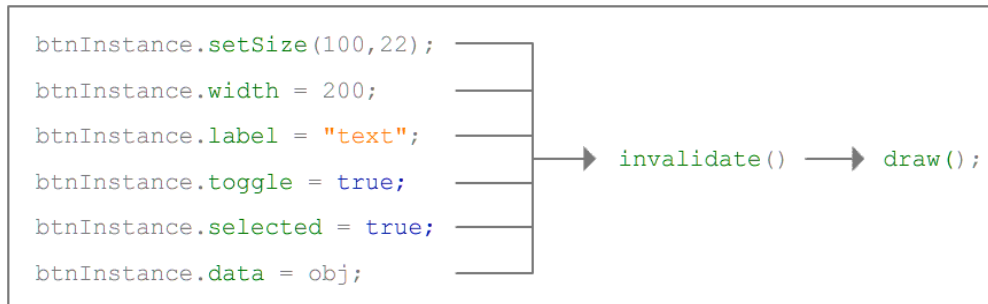
Wii™

CLIK

SWF

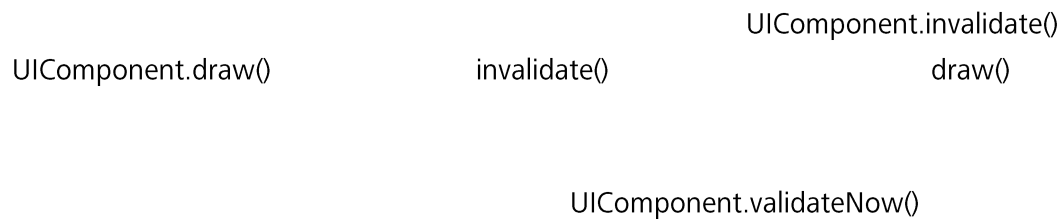
```
myButton.addEventListener("click", this, "onCursorClick");
function onCursorClick(event:Object):Void {
    switch (event.mouseIndex) {
        ...
    }
}
```

4.6



63: CLIK

4.6.1



4.7

Scaleform CLIK

1.

2.

scale9grid

Flash

scale9Grid

,

,

CLIK

,

scale9Grid

Scaleform

container

,

Flash/Flex

```

        CLIK
        ,
        container (
        ,
        ,
        Flash
        IDE
        Flash
        IDE
        MovieClips
        LivePrevi
        ews
        Scaleform
        Player
        CLIK

```

4.7.1 Scale9Grid

```

        Scaleform Player
        Scale9Grid
        MovieClip
        scale9grid
        Flash
        MovieClips
        Player
        Scaleform
        MovieClip
        scale9grid
        Scale
        9grid

```

4.7.2

```

Constraints (gfx.utils.Constraints)
        ScrollBar
        Constraints
        ScrollBar
        configUI()
        draw()
        draw()

```

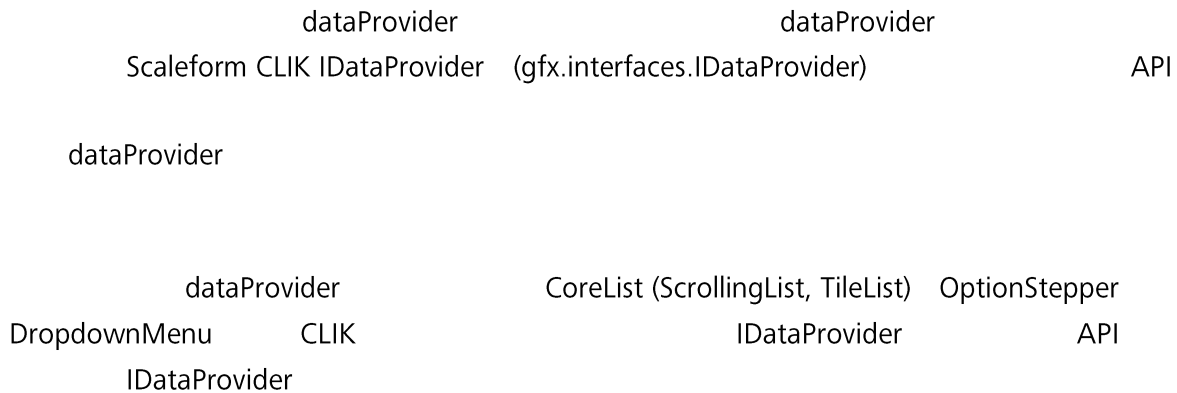
```

private function configUI():Void {
    ...
    constraints = new Constraints(this);
    //         upArrow
    constraints.addElement(downArrow, Constraints.BOTTOM);
    constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
    ...
}

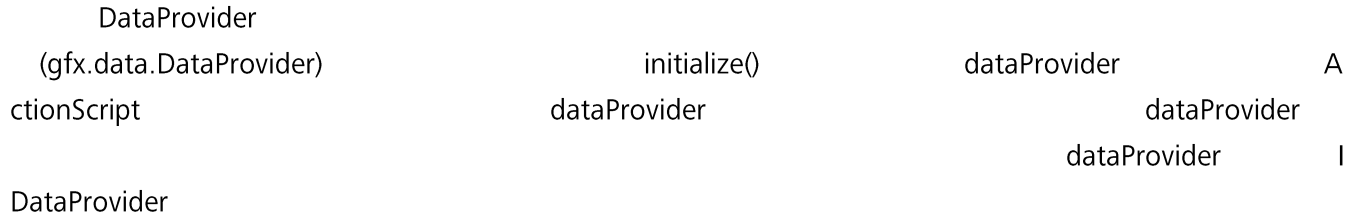
```

```
private function draw():Void {
    ...
    constraints.update(__width, __height);
    ...
}
```

12.



4.8.1



```
myComponent.dataProvider = [ "data1",
    4.3,
    {label:"anObjectElement", value:6} ];
```

dataProvider

- *length*:
- *requestItemAt*: dataProvider
OptionStepper
- *requestItemRange*: dataProvider
ScrollingList
- *indexOf*:
- *invalidate*: dataProvider
dataProvider
dataChange

dataProvider ExternalInterface.call

Scaleform CLIK

Flash 8

Scaleform

Tween

easing

mx.transiti

ons.easing.*

CLIK Tween

Tween

Flash 8

MovieClip

MovieClips

Tween

CLIK

CLIK

Tween onEnterFrame

MovieClip

Tween

Tween

Tween

MovieClip

Twelve

Twelve

MovieClip

Tweens

Tween MovieClip

```
tweenTo()
```

tweenFrom() tweenTo()

MovieClip

```
tweenFrom(
```

)

Tween

MovieClip

Tween

T

```
ween.init()
```

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();           // tween      MovieClip
//          1          Tween          alpha
myMovieClip.tweenTo(1, { x: 200, alpha: 0}, Strong.easeIn);
```

Tween

Tween

onTweenComplete()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
mc.onTweenComplete = function() {
    trace("Tween has finished!");
}
```

```
mc.tweenTo(5, {_rotation: 20}, Bounce.easeOut);
```

4.10

Scaleform

```
CLIK      PopUpManager  (gfx.managers.PopUpManager)
          PopUpManager
```

4.10.1

```

PopUpManager
    MovieClip
        CLIK
        ID
        createPopUp()
        MovieClips

```

```
import gfx.managers.PopUpManager;
PopUpManager.createPopUp(context, "MyToolTipLinkageID",
```

4.11.1



64: DragDemo



ItemSlot				
(gfx.controls.ItemSlot)	InventorySlot			
ItemSlot	DragTarget	CLIK		ItemSlot

4.12

4.12.1 Delegate

Delegate	(gfx.utils.Delegate)	Scaleform
CLIK	DragManager	

```
eventProvider.onMouseMove = Delegate.create(this, doDrag);
```

4.12.2 Locale

Locale	(gfx.utils.Locale)					
		Scaleform				ActionScript
			Button	Label	TextInput	
		Locale.getTranslatedString()				
						Locale

5

Scaleform CLIK

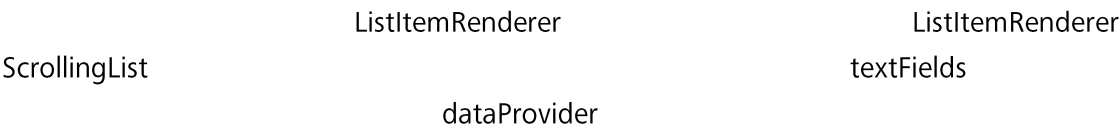
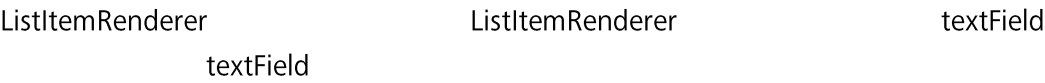
5.1

Scaleform CLIK

5.1.1 **textField ListItem Renderer**



65: **ScrollingList**



```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger",   lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger",   lname: "Woods"},
                     {fname: "Babe",    lname: "Ruth"},
                     {fname: "Wayne",   lname: "Gretzky"},
                     {fname: "Usain",   lname: "Bolt"}];
```



	ListItemRenderer	textField	textField	
	ListItemRenderer	MyItemRenderer		ListItem
mRenderer			MyItemRenderer.as	
FLA				

```
import gfx.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

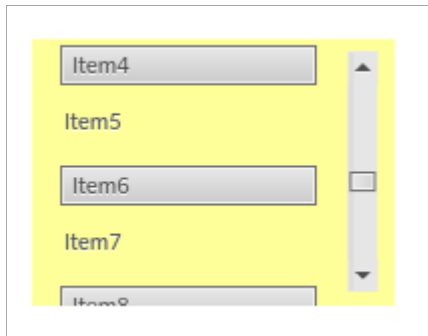
    public function MyItemRenderer() { super(); }

    public function setData(data:Object):Void {
        this.data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

    private function updateAfterStateChange():Void {
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}
```

ListItemRenderer	setData			
updateAfterStateChange		setData		(Scrollin
gList TileList)	ListItemRenderer	textField	MyItemRender	
er	textField		updateAfterStateChange	
	ListItemRenderer		textField	
		ListItemRenderer	dataProvider	
stItemRenderer	textField	textField1		Li
textField2		MyItemRenderer	MyItemRenderer	
	itemRenderer	ListItemRenderer	MyItemRenderer	
FLA			dataProvider	fname
lname				

5.1.2



66:

CLIK

CLIK ScrollBar

- **actions:** ActionScript
- **scrollbar:** CLIK ScrollBar sb
- **mask** MovieClip
- **content**
- **background**

content y 0

ActionScript

code

```
// 133
sb.setScrollProperties(1, 0, content._height - 133);
sb.position = 0;

sb.addEventListener("scroll", this, "onScroll");
function onScroll() {
    content._y = -sb.position;
}
```

setScrollProperties

Scaleform

HUD

Scaleform

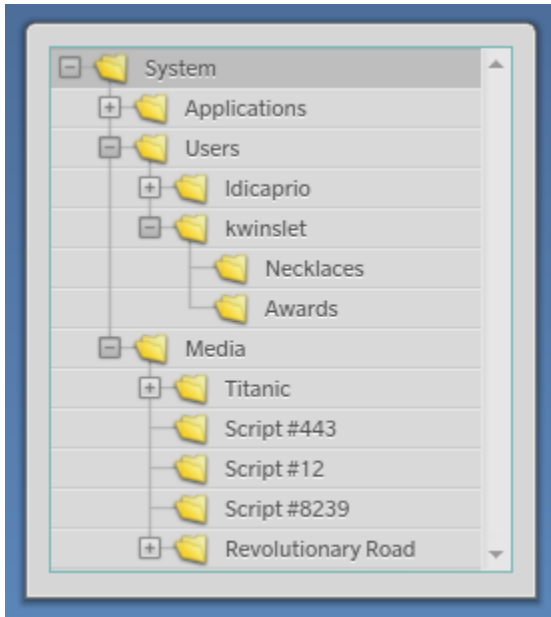
5.2

Scaleform

CLIK

Flash ActionScript 2

5.2.1 **ScrollingList TreeView**



67: TreeView

TreeView Demo

(Resources/AS2/CLIK/demos/TreeViewDemo.fla) ScrollingList

ListItemRenderer DataProvider

ScrollingList

-
-

tabbed

ListItemRenderer tab

•

ListItemRenderer

ListItemRenderer

Tre

eViewItemRenderer

TreeViewDataProvider

TreeViewConstants

Resources/AS2/CLIK/demos/com/

ActionScript

2

```
var root:Object =
{
    label: "System",
    nodes:[
        {label: "Applications",
        nodes:[
            {label: "CGStudio",
            nodes:[
                {label: "Data"},
                {label: "Samples"},
                {label: "Config"}
            ]
            },
            {label: "Money Manager"},
            {label: "Role Call v6"},
            {label: "Super Draft",
            nodes:[
                {label: "Templates"},
                ...
            ]
            }
        ]
    ]
}
```

label

nodes

ScrollingList

TreeViewDataProvider

ListItemRenderer

/

/

TreeViewDataProvider

CLIK

TreeViewItemRenderer ScrollingList TreeViewDataProvider ListItemRenderer

TreeViewItemRenderer tab ListItemRenderer tex

tField

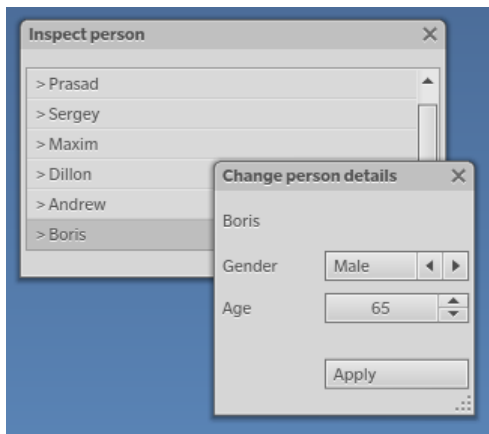
TreeViewConstant

- open node closed node leaf node;
-
-

ScrollingList ScrollBar

/ / ScrollingList

5.2.2



68:

Window Demo

(Resources/AS2/CLIK/demos/WindowDemo.fla)

Resources/AS2/CLIK/demos/com/scaleform/

-
-
-
-
-

Window

Title					
formPadding					
formType		symbol			
swf		symbol		swf SWF	
formSource		formType symbol SWF			
formType		swf			
allowResize					
minWidth, maxWidth, minHeight, maxHeight		0			

```
[Inspectable(name="formType", enumeration="symbol,swf")
private var _formType:String = "symbol";
```

Flash

_formTy

pe

formType

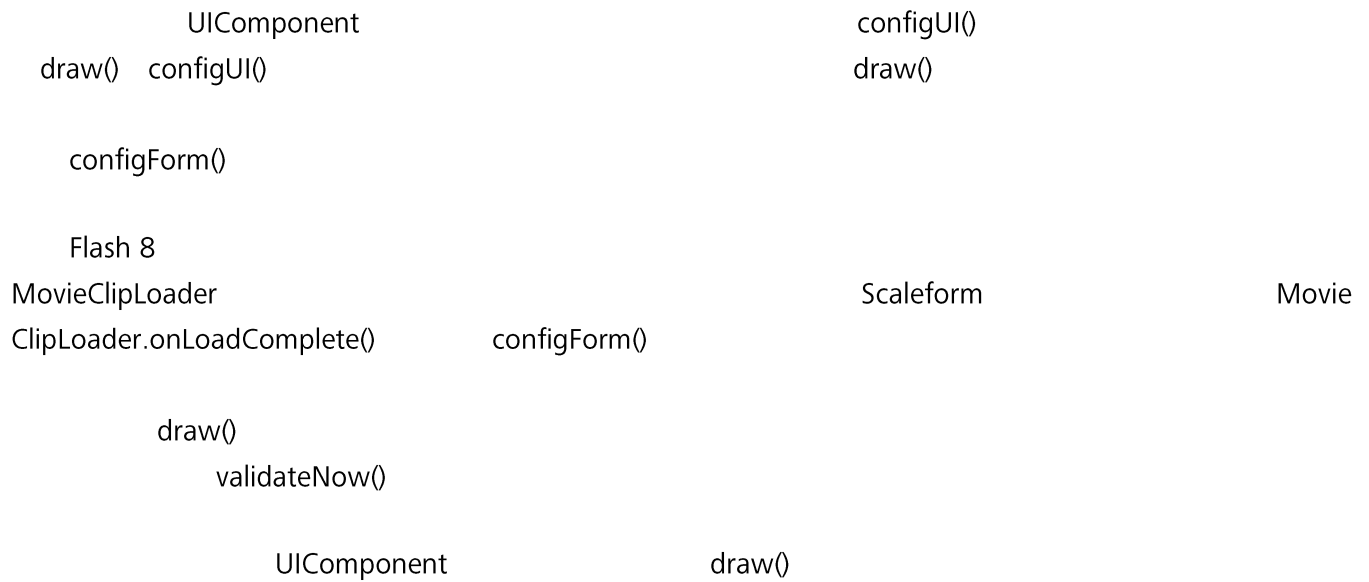
```
[Inspectable(defaultValue="Title")]
```

```

public function get title():String { return _title; }
public function set title(value:String):Void {
    _title = value;
    invalidate();
}

```

CLIK



dowManager

WindowManager

z Win